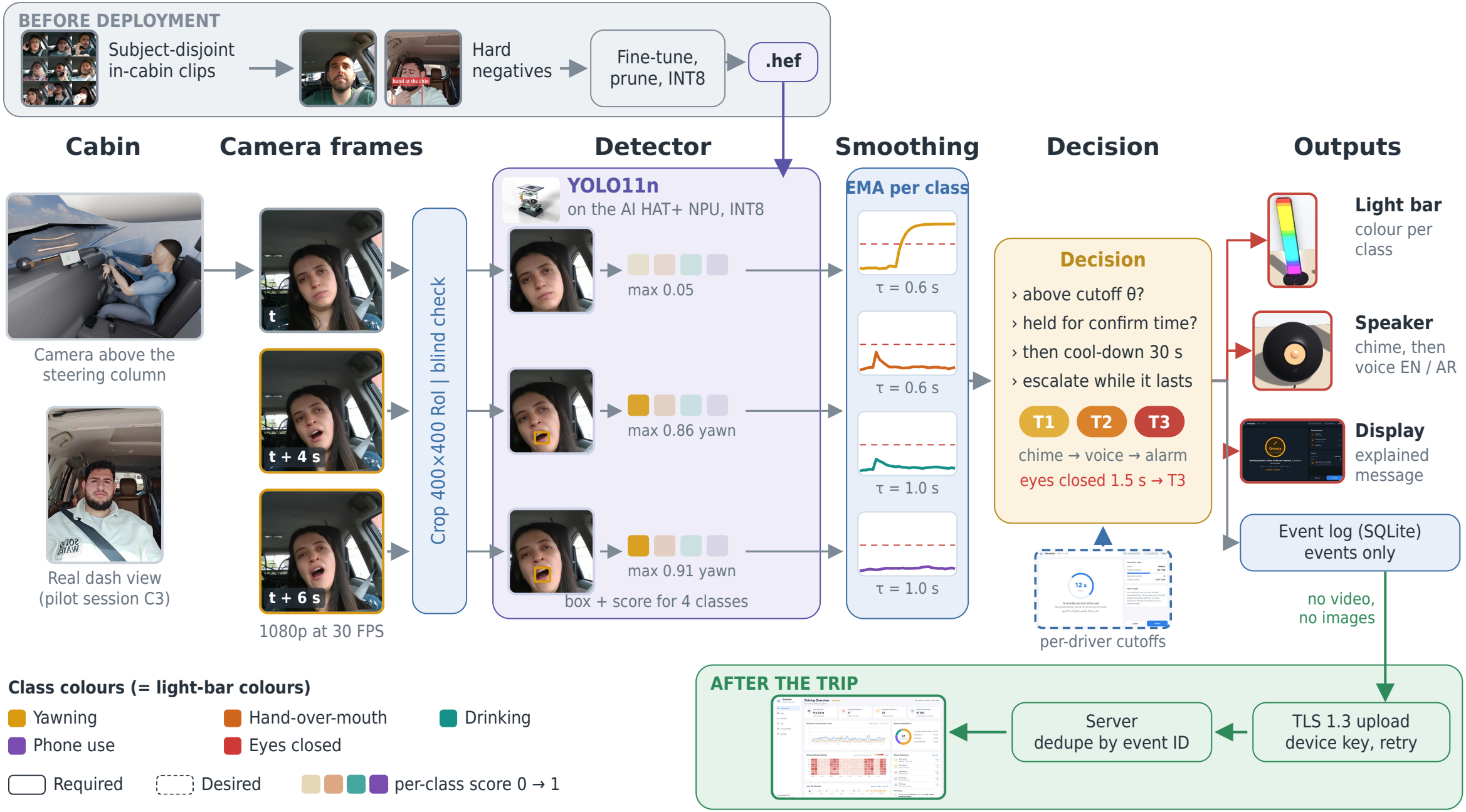

BEFORE DEPLOYMENT
Subject-disjoint in-cabin clips
Hard negatives
Fine-tune, prune, INT8
.hef
Cabin
Camera frames
Detector
Smoothing
Decision
Outputs
Camera above the steering column
Real dash view (pilot session C3)
t
t + 4 s
t + 6 s
1080p at 30 FPS
Crop 400×400 RoI | blind check
YOLO11n
on the AI HAT+ NPU, INT8
max 0.05
max 0.86 yawn
max 0.91 yawn
box + score for 4 classes
EMA per class
τ = 0.6 s
τ = 0.6 s
τ = 1.0 s
τ = 1.0 s
Decision
› above cutoff θ?
› held for confirm time?
› then cool-down 30 s
› escalate while it lasts
T1
T2
T3
chime → voice → alarm
eyes closed 1.5 s → T3
per-driver cutoffs
Light bar
colour per class
Speaker
chime, then voice EN / AR
Display
explained message
Event log (SQLite)
events only
no video,
no images
AFTER THE TRIP
TLS 1.3 upload
device key, retry
Server
dedupe by event ID
Class colours (= light-bar colours)
Yawning
Hand-over-mouth
Drinking
Phone use
Eyes closed
Required
Desired
per-class score 0 → 1